9 Discrete Mathematics: Combinatory (18)

9.0.1 UGC NET CSE | December 2012 | Part 3 | Question: 25

The number of distinct bracelets of five beads made up of red, blue and green beads (two bracelets are indistinguishable if the rotation of one yield another) is,

A. 243 B. 81 C. 51 D. 47

ugcnetcse-dec2012-paper3

Answer key

9.1 Binary Tree (1)

9.1.1 Binary Tree: UGC NET CSE | July 2016 | Part 3 | Question: 31

The number of different binary trees with 6 nodes is

A. 6 B. 42 C. 132 D. 256

ugcnetcse-july2016-paper3 combinatory binary-tree

Answer key

9.2 Combinatory (7)

9.2.1 Combinatory: UGC NET CSE | December 2012 | Part 3 | Question: 20

The number of ways to distribute n distinguishable objects into k distinguishable boxes, so that n_i objects are placed into box i, $i = 1, 2, \dots k$ equals which of the following?

A. $\frac{n!}{n_1!+n_2!+\cdots+n_k!}$

B. $\frac{n_1!+n_2!+\cdots+n_k!}{n_1!n_2!\dots n_k!}$

C. $\frac{n_1!}{n_1!n_2!\dots n_k!}$

D. $\frac{n_1!n_2!\dots n_k!}{n_1!-n_2!-n_3!\cdots-n_k!}$

ugcnetcse-dec2012-paper3 discrete-mathematics combinatory

Answer key

9.2.2 Combinatory: UGC NET CSE | December 2012 | Part 3 | Question: 29

58 lamps are to be connected to a single electric outlet by using an extension board each of which has four outlets. The number of extension boards needed to connect all the light is

A. 28 B. 29 C. 20 D. 19

ugcnetcse-dec2012-paper3 combinatory

Answer key

9.2.3 Combinatory: UGC NET CSE | December 2013 | Part 2 | Question: 38

If n and r are non-negative integers and n $\geq$ r, then p(n+1, r) equals to

A. $\frac{P(n,r)(n+1)}{(n+1-r)}$

B. $\frac{P(n,r)(n+1)}{(n-1+r)}$

C. $\frac{P(n,r)(n-1)}{(n+1-r)}$

D. $\frac{P(n,r)(n+1)}{(n+1+r)}$

ugcnetcse-dec2013-paper2 discrete-mathematics combinatory

Answer key

9.2.4 Combinatory: UGC NET CSE | December 2015 | Part 2 | Question: 1

How many committees of five people can be chosen from 20 men and 12 women such that each committee contains at least three women

A. 75240 B. 52492 C. 41800 D. 9900

combinatory ugcnetcse-dec2015-paper2

Answer key

9.2.5 Combinatory: UGC NET CSE | June 2014 | Part 2 | Question: 25

How many cards must be chosen from a deck to guarantee that at least

i. two aces of two kinds are chosen.
ii. two aces are chosen.
iii. two cards of the same kind are chosen.
iv. two cards of two different kinds are chosen.

A. $50, 50, 14, 5$
B. $51, 51, 15, 7$
C. $52, 52, 14, 5$
D. $51, 51, 14, 5$

ugcnetcse-june2014-paper2 combinatory

Answer key

9.2.6 Combinatory: UGC NET CSE | June 2019 | Part 2 | Question: 2

How many ways are there to place 8 indistinguishable balls into four distinguishable bins?

A. 70 B. 165 C. ${}^8C_4$ D. ${}^8P_4$

ugcnetcse-june2019-paper2 combinatory

Answer key

9.2.7 Combinatory: UGC NET CSE | October 2020 | Part 2 | Question: 2

How many ways are there to pack six copies of the same book into four identical boxes, where a box can contain as many as six books?

A. 4 B. 6 C. 7 D. 9

ugcnetcse-oct2020-paper2 discrete-mathematics combinatory

Answer key

9.3 Counting (3)

9.3.1 Counting: UGC NET CSE | December 2018 | Part 2 | Question: 33

The number of substrings that can be formed from string given by

$a\,d\,e\,f\,b\,g\,h\,n\,m\,p$

is

A. 10 B. 45 C. 55 D. 56

ugcnetcse-dec2018-paper2 combinatory counting

Answer key

9.3.2 Counting: UGC NET CSE | Junet 2015 | Part 2 | Question: 1

How many strings of 5 digits have the property that the sum of their digits is 7?

A. 66 B. 330 C. 495 D. 99

ugcnetcse-june2015-paper2 discrete-mathematics counting

Answer key

9.3.3 Counting: UGC NET CSE | Junet 2015 | Part 2 | Question: 3

In how many ways can 15 indistinguishable fish be placed into 5 different ponds, so that each pond contains at least one fish?

A. 1001 B. 3876 C. 775 D. 200

ugcnetcse-june2015-paper2 combinatory counting

Answer key ☟

9.4 Euler Phi Function (1)

9.4.1 Euler Phi Function: UGC NET CSE | June 2019 | Part 2 | Question: 63

Consider the Euler's phi function given by

$$\phi(n) = n \prod_{p/n} \left(1 - \frac{1}{p}\right)$$

where p runs over all the primes dividing n. What is the value of $\phi(45)$?

A. 3 B. 12 C. 6 D. 24

ugcnetcse-june2019-paper2 euler-phi-function

Answer key ☟

9.5 Inclusion Exclusion (1)

9.5.1 Inclusion Exclusion: UGC NET CSE | June 2019 | Part 2 | Question: 3

How many bit strings of length ten either start with a 1 bit or end with two bits 00?

A. 320 B. 480 C. 640 D. 768

ugcnetcse-june2019-paper2 combinatory inclusion-exclusion

Answer key ☟

9.6 Parenthesization (1)

9.6.1 Parenthesization: UGC NET CSE | July 2016 | Part 2 | Question: 25

In how many ways can the string $A \cap B - A \cap B - A$ be fully paranthesized to yield an infix expression?

A. 15 B. 14 C. 13 D. 12

ugcnetcse-july2016-paper2 parenthesization combinatory

Answer key ☟

9.7 Permutation and Combination (1)

9.7.1 Permutation and Combination: Computer Science - UGC NET 2021 [Question ID = 2353]

How many ways are there to assign 5 different jobs to 4 different employees if every employee is assigned at least 1 job ?

1. 1024
2. 625
3. 240
4. 20

discrete-mathematics permutation-and-combination engineering-mathematics

Answer key ☟

9.8 Pigeonhole Principle (1)

9.8.1 Pigeonhole Principle: UGC NET CSE | June 2019 | Part 2 | Question: 7

How many cards must be selected from a standard deck of 52 cards to guarantee that at least three hearts are present among them?

A. 9 B. 13 C. 17 D. 42

ugcnetcse-june2019-paper2 combinatory pigeonhole-principle

Answer key ☟

9.9 Recurrence Relation (1)

9.9.1 Recurrence Relation: UGC NET CSE | December 2012 | Part 2 | Question: 10

The number of bit strings of length eight that will either start with a 1 bit and end with two bits 00 shall be

A. 32 B. 64 C. 128 D. 160

ugcnetcse-dec2012-paper2 combinatory recurrence-relation

Answer key ☟

Answer Keys

9.0.1	C	9.1.1	C	9.2.1	C	9.2.2	D	9.2.3	A
9.2.4	B	9.2.5	A	9.2.6	B	9.2.7	D	9.3.1	D
9.3.2	B	9.3.3	A	9.4.1	D	9.5.1	C	9.6.1	B
9.7.1	Q-Q	9.8.1	D	9.9.1	D				